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NMAM INSTITUTE OF TECHNOLOGY, NITTE**Off-Campus Centre of Nitte (Deemed to be University)****Third Semester B.Tech (Mechanical) (CBCS) Degree Examinations**

December 2023

ME1006-1 – MANUFACTURING PROCESSES

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which type of sand molding machine is most suitable for producing large castings?
A) Jolt-squeeze molding machine B) Flaskless molding machine
C) Vacuum molding machine D) Sand slinger machine
2. The life of a pattern is most likely to depend upon which of the following factor?
A) Number of castings produced B) Type of cooling rate of the casting
C) Size of the casting D) Size of the pattern
3. What is the function of cores in the casting process?
A) To support the pattern B) To provide differential cooling rates at specific portions
C) To make holes and cavities D) For ease of flow of the molten material
4. A draft allowance is provided to
A) All linear faces B) Only the interior dimensions
C) Only the exterior dimensions D) Only the dimensions that are perpendicular to the parting plane
5. Which type of centrifugal casting method is used to produce 'non-symmetrical' shaped castings?
A) True centrifugal casting B) Semi centrifugal casting
C) Centrifuging D) Non centrifugal casting
6. Which of the following synthetic resins are used for mixing with sand in shell moulding?
A) Thermosetting resins B) Epoxy resins
C) Fiber glass resins D) Kevlar
7. Which of the following metals is used for the preparation of patterns in shell moulding?
A) Aluminium B) Grey cast iron
C) Copper D) Brass
8. Which of the following is used as the pattern material in precision investment casting?
A) Metal B) Wax
C) Plastic D) Wood
9. In two high rolling mill, what is the direction of rolling of the two rollers?
A) Clockwise-anticlockwise B) Clockwise-clockwise
C) Anticlockwise-Anticlockwise D) Stationery-clockwise
10. Which of the following is not made using extrusion?
A) Window frames B) Doors
C) Helmets D) Rods
11. Which of the following dies is not the type of cutting dies in the metal cutting operations?
A) Drawing dies B) Blanking dies
C) Piercing dies D) Notching dies
12. What is the role of lubricants in the powder metallurgy process?
A) To reduce friction between metal particles B) To increase the density of the final product
C) To reduce the cost of production D) To increase the strength of the final product

13. Glass tubes are made using
 A) Extrusion process B) Danner process
 C) Spinning process D) Casting process
14. The heat treatment process of glass which leads to sharp edges when broken is
 A) Tempering B) Annealing
 C) Quenching D) Sintering
15. _____ screw is used in polymer extrusion process.
 A) Archimedes B) Newtons
 C) Cast iron D) Aluminium
16. In blow molding process, the intermediate tube is called
 A) Carson B) Parison
 C) Harrison D) U-tube
17. Oxy-acetylene process is most suited for which of the following process of joining?
 A) Metal wires B) Metal sheets
 C) Metal tubes D) Metal bars
18. Which kind of resistance is experienced in upset butt welding?
 A) Electric resistance B) Magnetic resistance
 C) Thermal resistance D) Air resistance
19. What is the main advantage of TIG welding over other welding techniques?
 A) High welding speed B) Ability to weld thick materials
 C) High precision and control D) Low cost of equipment
20. Friction welding is a type of _____ welding.
 A) Fusion B) Solid state
 C) Gas D) Laser

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Write a note on different types of cores used in casting process.	8	L1	1	1
	b) Explain with sketch the continuous casting process.	8	L2	2	1
2.	a) Explain with sketch the working of jolt squeeze machine and sand slinger.	8	L2	1	1
	b) Explain with sketch the hot and cold pressure die casting processes.	8	L2	2	1
3.	a) What are the ingredients of molding sand? Write a note on its desirable properties.	8	L1	1	1
	b) What is 'lost-wax casting'? Explain the process with schematic drawings and highlight three reasons to opt this production method.	8	L2	2	1
Unit – II					
4.	a) Compare the process of direct and indirect extrusion with schematic sketches.	8	L2	3	1
	b) What are the desirable properties of ceramic materials that make us go for it? What are the steps involved in their processing?	8	L1	4	1
5.	a) Write a note on following forming processes: i) Shearing ii) Tube drawing	8	L1	3	1
	b) Describe the process of glass annealing. Mention the unique properties of glass material.	8	L1	4	1
6.	a) Narrate the functioning of progressive die set up and explain when this setup is preferred over others.	8	L2	3	1
	b) With a schematic representation, explain the Blow moulding technique for plastic materials.	8	L2	4	1

Unit – III

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| 7. a) | Write the causes for following welding defects: | | | | |
| | i) Cracks | | | | |
| | ii) Porosity | | | | |
| | iii) Undercuts | | | | |
| | iv) Distortion | 8 | L1 | 5 | 1 |
| b) | Describe the principle of Gas welding mentioning the chemistry of the flame formation. Write the nature of following types of flames obtained from gas welding torch: | | | | |
| | i) Carburizing flame | | | | |
| | ii) Oxidizing flame | 8 | L1 | 5 | 1 |
| 8. a) | What is the need for welding? How it is different from brazing and soldering processes? Classify welding process. | 8 | L2 | 5 | 1 |
| b) | Explain with sketch friction welding and ultrasonic welding processes. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
